COMPILING THE ALMGREN–PITTS COMBINATORIAL ARGUMENT

XINZE LI

Abstract. The Almgren–Pitts combinatorial argument — the cover-refinement plus bookkeeping backbone of Step 2 in min-max constructions of minimal hypersurfaces — is a recurring technical step that each author re-derives in local notation. We extract it from the literature as a standalone metric-combinatorial algorithm and formalize it in Lean 4 as the importable library `CombArg`.

1. Introduction

A minimal hypersurface is a critical point of the area functional in codimension one. On a closed Riemannian manifold (M^{n+1}, g), direct minimization over a fixed homology class may produce only degenerate configurations, and the relevant critical points are typically of saddle type. The resolution, following Birkhoff (1917) for closed geodesics and Almgren–Pitts for higher-dimensional varifolds, is a *min-max* construction; regularity of the output is Schoen–Simon [SS81].

Theorem 1.1 (Almgren–Pitts min-max theorem [Pit81, SS81])**.** *Every closed Riemannian manifold (M^{n+1}, g) admits a nontrivial embedded minimal hypersurface $\Sigma \subset M$ whose singular set* $\operatorname{Sing}\Sigma$ *has Hausdorff dimension at most $n-7$.*

The core technical result of Pitts's proof — used in every subsequent treatment [CDL03, DLT13, DLR18, DL16, MN14] — is the Almgren–Pitts combinatorial argument, stated in the Colding–De Lellis formulation as follows.

Proposition 1.2 (Almgren–Pitts combinatorial Argument, [CDL03, Prop. 3.4])**.** *Let Λ be a homotopically closed family of sweepouts of M^{n+1}, and $m_0 = \inf_\Lambda \mathcal{F}$. There exists a min-max sequence $\{\Gamma^N\}$ converging as varifolds to a stationary integral varifold of mass m_0, with Γ^N $1/N$-almost-minimizing in every pair $(U^1, U^2) \in \mathcal{CO}$ of $\mathcal{H}^n$-disjoint open sets, for all N sufficiently large.*

The proof of Proposition 1.2 reduces to a one-parameter claim on the sweepout [DLT13, §3.2]:

> **Claim.** For each N sufficiently large, there exists $t_N \in [0,1]$ such that $\Gamma^N := \Gamma^N_{t_N}$ is $1/N$-almost-minimizing in every $(U^1, U^2) \in \mathcal{CO}$ and $\mathcal{H}^n(\Gamma^N) \geq m_0 - 1/N$.

The Claim is proved by contradiction: from its negation, every t with $\mathcal{H}^n(\partial\Omega^N_t) \geq m_0 - 1/N$ admits a *local replacement* with saving $1/(4N)$ [DLT13, Lemma 3.1]; a combinatorial cover-refinement and bookkeeping argument on $[0,1]$ ([DLT13, §3.2] Steps 1–2) then assembles from these local replacements a modified sweepout with supremum strictly below m_0, contradicting $m_0 = \inf_\Lambda \mathcal{F}$. The combinatorial argument depends on no geometric measure theory, no admissible class, and no ambient regularity; it is a bookkeeping construction on continuous real-valued functions, yet has no canonical reference.

Stripped of its GMT instantiation, the content of Steps 1–2 is the following scalar statement, which is the precise theorem we formalize.

Theorem 1.3 (Extracted scalar version of Steps 1–2)**.** *Let X be a pseudo-metric space with an abstract paired-cover structure. Let $f : [0,1] \to \mathbb{R}$ be continuous with $m_0 = \sup f$ and $m_0 > 0$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose that at every $t \in [0,1]$ with $f(t) \geq m_0 - 1/N$ there is a* local

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Correspondence: `lixinze@math.utoronto.ca`.

witness*: an open neighborhood $U_t \ni t$, a paired cover c_t in X, and a continuous replacement energy $E_t : [0,1] \to \mathbb{R}$ satisfying $f(s) - E_t(s) \geq 1/(4N)$ for every $s \in U_t$. Then there exists $f' : [0,1] \to \mathbb{R}$ with*

$$\sup f' \;\leq\; m_0 - \frac{1}{4N}.$$

The boundary $\partial\Omega_t$ is replaced by a continuous $f : [0,1] \to \mathbb{R}$; the Hausdorff-measure hypothesis becomes $m_0 = \sup f$; the nested pair (A_t, B_t, U_t') collapses to a single neighborhood U_t; the replacement family $\{\partial\Omega_{t,\tau}\}$ collapses to a single continuous E_t; the almost-minimizing hypothesis disappears. What remains is the combinatorial bookkeeping. Theorem 1.3 reappears in §2.3 as the library's public one-parameter API (Theorem 2.8); it is a specialization of an abstract core (Theorem 2.7) parametric in the sweepout space and in a pair (δ, ε) replacing the hardcoded $(1/N, 1/(4N))$.

Scope. This note formalizes Steps 1–2 of the proof of the Claim as a Lean 4 library, `CombArg`. We do *not* formalize [DLT13, Lemma 3.1] (the replacement family), the $(U^1, U^2) \in \mathcal{CO}$ disjoint-pair structure, or the final contradiction against $m_0 = \inf_\Lambda \mathcal{F}$; these depend on the GMT and admissible-class framework of the ambient construction and remain the consumer's responsibility. The library ships a single abstract theorem (§2.2) parametric in the sweepout space together with a one-parameter specialization to $[0,1]$ (§2.3); the formalization is machine-verified against `Mathlib`, declares no custom axioms, and contains no `sorry`. Source: https://github.com/MathNetwork/comb_arg.

2. Specification

We present the library specification following standard software documentation conventions: interface first (§2.1), public API (§2.2–§2.3), internal lemmas (§2.4).

2.1. Interface: types and structures.

Throughout, `K` denotes a topological space (the parameter space), `X` a pseudo-metric space (the ambient space whose covers witness replacements), and `f : K → ℝ` a continuous scalar energy.

Definition 2.1 (`PairableCover` class). [✓ `PairableCover`]

A `PairableCover X` instance declares, for each abstract `c : Cover`, a left region and a right region in X and a scalar diameter, subject to two axioms: the two regions are disjoint, and overlapping combined regions of diameter-comparable covers are ordered by inclusion.

Definition 2.2 (`LocalWitness` structure). [✓ `LocalWitness`]

A `LocalWitness K X f t` ε carries an open neighborhood $U \ni t$ in K, a cover `c : Cover`, a continuous replacement energy $E : K \to \mathbb{R}$, and the inequality $\forall s \in U,\ f(s) - E(s) \geq \varepsilon$.

Definition 2.3 (Near-critical set). [✓ `nearCritical`]

For continuous $f : K \to \mathbb{R}$, scalar m_0, and $\delta \geq 0$, the δ-near-critical set is $\{t \in K : f(t) \geq m_0 - \delta\}$. When $\delta = 1/N$ we write NC.

Definition 2.4 (`FiniteCoverWithWitnesses` structure). [✓ `FiniteCoverWithWitnesses`]

A `FiniteCoverWithWitnesses K f m₀` δ ε carries a finite nonempty index type ι, pieces $\{p_l \subseteq K\}_{l \in \iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \in \mathbb{R}_{>0}\}$, subject to four conditions: (I) $f - E_l \geq s_l$ on p_l; (II) each $t \in K$ lies in at most two pieces; (III) $s_l \geq \varepsilon$ for all l; (IV) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$. This is the abstract form of the refined covering $\{\overline{J_l}\}$ produced at the end of [DLT13, §3.2 Step 1]; conditions (II)–(III) are DLT's two-fold overlap and per-piece saving invariants.

Definition 2.5 (`InitialCover` structure). [✓ `InitialCover`]

Specialized to $K = \mathbf{I}$. An `InitialCover X f m₀ N` carries a finite family of open intervals $I_i = (c_i - r_i, c_i + r_i) \cap \mathbf{I}$ for $i = 1, \ldots, n$, each paired with a witness center $w_i \in$ NC and a local

witness at w_i with saving $1/(4N)$, subject to (a) spacing: $c_i + r_i < c_{i+2} - r_{i+2}$ for all valid i; (b) containment: $I_i \subseteq U_{w_i}$; (c) coverage: $\mathrm{NC} \subseteq \bigcup_i I_i$.[1] Corresponds to the initial covering $\{I_i\}$ of NC entering [DLT13, §3.2] *before* the Step 1 refinement.

Definition 2.6 (`PartialRefinement` structure). [✓ `PartialRefinement`]

Given `ic : InitialCover` and $\ell \in \mathbb{N}$, a `PartialRefinement ic` ℓ carries pieces $J_1, \ldots, J_\ell \subseteq \mathbf{I}$ and an index assignment $\sigma : \{1, \ldots, \ell\} \to \{1, \ldots, n\}$ with (i) $J_k \subseteq I_{\sigma(k)}$ for each k; (ii) $I_{\sigma(k)} \subseteq \bigcup_{k'} J_{k'}$ for each k. This captures the *mid-induction state* of [DLT13, §3.2 Step 1], where each I_j has been either kept whole ("the same") or split ("into two halves").

2.2. Public API: abstract core.

Theorem 2.7 (Abstract core). [✓ `exists_sup_reduction_of_cover`]
Input. *A compact, nonempty topological space K; a continuous $f : K \to \mathbb{R}$; $m_0 = \sup f$; scalars $0 < \varepsilon \le \delta$; a `FiniteCoverWithWitnesses`$(K, f, m_0, \delta, \varepsilon)$ (Definition 2.4).*
Output. *There exists $f' : K \to \mathbb{R}$ with $\sup f' \le m_0 - \varepsilon$.*

2.3. Public API: one-parameter application.

Theorem 2.8 (One-parameter application). [✓ `exists_sup_reduction`]
Input. *A pseudo-metric space X with a `PairableCover X` instance (Definition 2.1); a continuous $f : \mathbf{I} \to \mathbb{R}$; $m_0 \in \mathbb{R}$ with $m_0 > 0$ and $m_0 = \sup f$; $N \in \mathbb{N}$ with $N > 0$; for every $t \in \mathrm{NC}$ a `LocalWitness` at t with saving $1/(4N)$ (Definition 2.2).*
Output. *There exists $f' : \mathbf{I} \to \mathbb{R}$ with $\sup f' \le m_0 - 1/(4N)$.*
Specializes Theorem 2.7 to $K = \mathbf{I}$, $\delta = 1/N$, $\varepsilon = 1/(4N)$.

2.4. Internal lemmas.

The declarations below are consumed by Theorem 2.8; proofs are sketched in §3.

Proposition 2.9 (Compactness of NC). [✓ `isCompact_nearCritical`, ✓ `nearCritical_nonempty`]

If K is compact and f continuous, NC *is compact. If additionally $m_0 = \sup f$ and $N > 0$,* NC *is nonempty.*

- ✓ `exists_initialCover` — existence of an `InitialCover` via a grid plus Lebesgue-number construction on the compact sublevel set NC.
- ✓ `exists_terminal_refinement` — existence of a terminal `PartialRefinement` with injective σ via bounded smallest-index induction with two-case dispatch.
- ✓ `not_three_overlap` — at most two among any three distinct `InitialCover` intervals overlap at a common point (parity rescue).
- ✓ `saving_bound_closure` — the open-neighborhood saving bound in `LocalWitness` extends to the closure via continuity of E.
- ✓ `exists_refinement` — packages the 1D construction output as a `FiniteCoverWithWitnesses` with $(\delta, \varepsilon) = (1/N, 1/(4N))$ on $\mathbf{I}$; the input to Theorem 2.7.

Figure 1 shows the layered architecture.

3. Proof structure

We split the proof of Theorem 2.8 into three stages, each one mirroring a segment of [DLT13, §3.2]:

Our stage	DLT13 §3.2 segment	Output
Stage A	(setup, pre-Step 1)	`InitialCover`
Stage B	Step 1 (refinement of the covering)	terminal `PartialRefinement`
Stage C	Step 2 (conclusion: bookkeeping)	`FiniteCoverWithWitnesses`

[1] Conditions (a) and (b) transcribe [DLT13, §3.2]'s spacing and subset conditions; the pair (c_i, w_i) separates interval centers from witness centers, required because grid points need not themselves be near-critical.

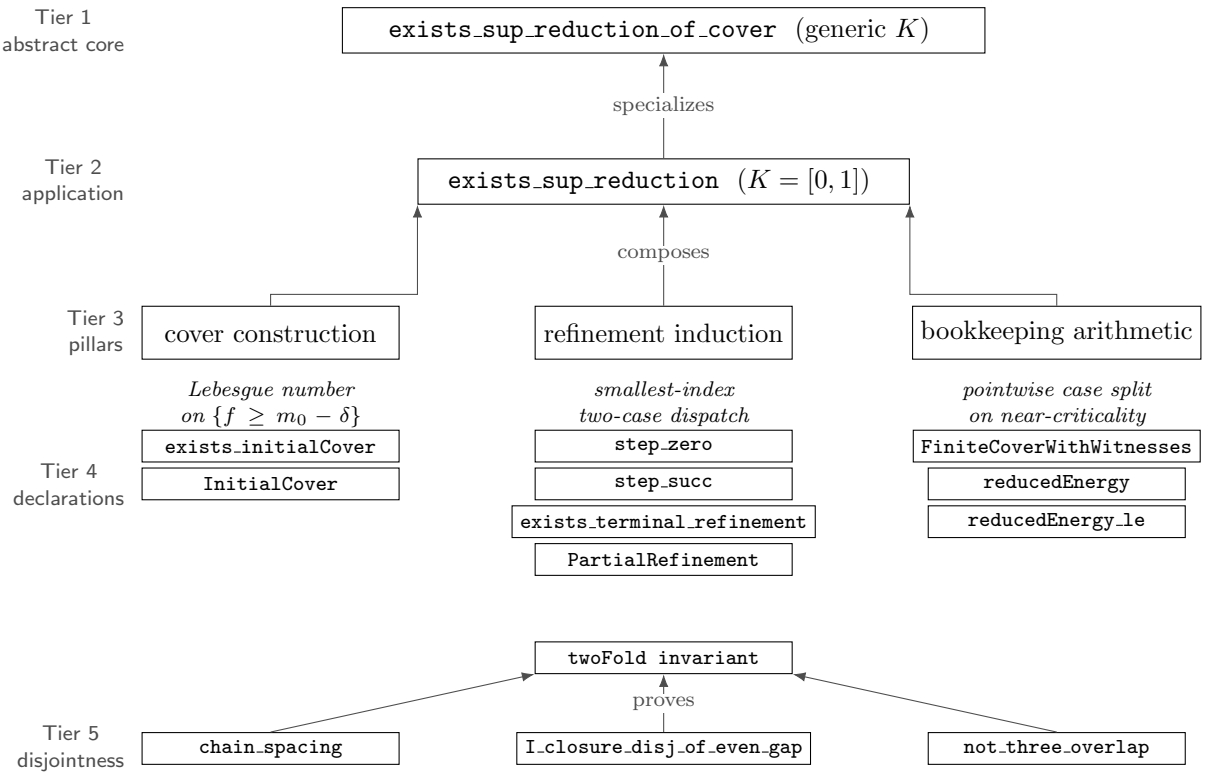

FIGURE 1. `CombArg` v0.1 architecture. The abstract core (Tier 1) is parametric in the sweepout space; the one-parameter application (Tier 2) instantiates $K = [0, 1]$ and composes three construction pillars (Tier 3). Each pillar's key declarations appear in Tier 4. The disjointness sub-module (Tier 5) supplies the two-fold overlap invariant consumed by bookkeeping.

Theorem 2.7 then consumes the output of Stage C and closes the chain.

3.1. Stage A: existence of the initial cover.

Proposition 3.1 (Existence of an initial cover). [✓ `exists_initialCover`]
Under the hypotheses of Theorem 2.8, an initial cover (Definition 2.5) exists.

Construction. Compactness of NC (Proposition 2.9) plus the open cover $\{U_t\}_{t \in \mathrm{NC}}$ afforded by the witness hypothesis admit a Lebesgue number $\lambda > 0$: every λ-ball in NC fits inside some U_t. Choose $M \in \mathbb{N}$ with $M > 4/\lambda$, form the grid $c_k = k/M$ for $k = 0, \ldots, M$, and select those c_k within $\lambda/4$ of NC. For each selected c_k pick a witness center $w_k \in \mathrm{NC}$ within $\lambda/4$ of c_k; the ball of radius λ around w_k fits inside some U_{w_k}, hence so does the interval $I_k := (c_k - 2/(3M), c_k + 2/(3M)) \cap \mathbf{I}$. Spacing condition (a) holds because consecutive selected grid indices differ by at least 2 in the selected finset, so their centers differ by at least $2/M > 4/(3M) = 2 \cdot (2/(3M))$. Coverage condition (c) holds because every $t \in \mathrm{NC}$ is within $1/(2M) < 2/(3M)$ of its rounded grid neighbor, which is selected. Condition (b) holds by the Lebesgue argument. □

3.2. Stage B: the refinement induction.

This stage realizes [DLT13, §3.2 Step 1]: each initial interval I_j with $j \geq 2$ is either kept whole ("remains the same") or split ("into two halves"), depending on whether the paired cover at $t_{\sigma(\ell)}$ is compatible with the new t_{i^*}. Figure 2 reproduces the [DLT13, Fig. 1] schematic.

Formally, $\{J_l\}$ is built by induction from step_zero (base case) via step_succ_at (inductive step).

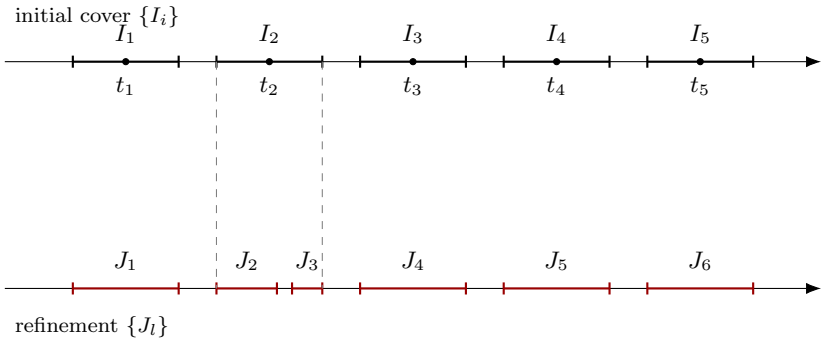

FIGURE 2. Refinement of the covering on $[0,1]$, after [DLT13, Fig. 1]. Top: the initial cover $\{I_i\}$ with witness centers $t_i \in \mathrm{NC}$. Bottom: the refined cover $\{J_l\}$; here I_2 is split into $J_2 \cup J_3$ because the paired cover at t_3 intersects the one at t_2, while I_1, I_3, I_4, I_5 are kept whole. In the resulting cover every $t \in [0,1]$ lies in at most two of the $\overline{J_l}$ (two-fold invariant).

Definition 3.2 (Base case)**.** [✓ `step_zero`]
step_zero(ic) is the length-1 partial refinement with $J_1 := I_1$ and $\sigma(1) := 1$.

Definition 3.3 (Inductive step)**.** [✓ `step_succ_at`]
Given a partial refinement pr of length $\ell + 1$ and an index $i^* \in \{1, \dots, n\}$ with $I_{i^*} \not\subseteq \bigcup_k J_k$, define a length-$(\ell + 2)$ extension by case split on whether $I_{i^*} \subseteq J_\ell \cup I_{\sigma(\ell)}$:

(Case 1) If yes, set $J_{\ell+1} := I_{i^*}$.
(Case 2) Otherwise, set $J_{\ell+1} := I_{i^*} \setminus I_{\sigma(\ell)}$.

In both cases set $\sigma(\ell + 1) := i^*$. Both invariants of Definition 2.6 are preserved; in Case 2 the preservation of (ii) uses the induction hypothesis on $I_{\sigma(\ell)} \subseteq \bigcup_k J_k$.[2]

Proposition 3.4 (Termination)**.** [✓ `exists_terminal_refinement`]
Starting from step_zero(ic) *and iterating* step_succ_at*, one reaches in at most n steps a partial refinement of some length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i.*

Argument. Let rem(pr) be the finite set of indices i with $I_i \not\subseteq \bigcup_k J_k$. One step of step_succ_at at $i^* \in \mathrm{rem}(\mathrm{pr})$ (chosen as the smallest such) produces pr′ with $i^* \notin \mathrm{rem}(\mathrm{pr}')$ and $\mathrm{rem}(\mathrm{pr}') \subseteq \mathrm{rem}(\mathrm{pr}) \setminus \{i^*\}$, hence strict decrease. Injectivity of σ follows from the case split in Definition 3.3: the new $\sigma(\ell + 1) = i^*$ is fresh because $i^* \in \mathrm{rem}$ while existing $\sigma(k)$ correspond to already-covered indices. □

3.3. **Stage C: assembly.** This stage realizes [DLT13, §3.2 Step 2]: from the refined $\{J_l\}$ together with the local-witness saving bound one packages a `FiniteCoverWithWitnesses` (Definition 2.4), whose three invariants (per-piece saving floor, two-fold overlap, coverage of NC) are precisely the data DLT's Step 2 bookkeeping consumes. Here we extend the saving bound from open J_k to $\overline{J_k}$ and derive the two-fold invariant from spacing (a). Theorem 2.7 then turns the assembled cover into the sup-reducing competitor f' (our formalization stops here; DLT proceeds to derive a contradiction against $m_0 = \inf_\Lambda \mathcal{F}$, which is the caller's responsibility).

Proposition 3.5 (TwoFold via parity)**.** [✓ `terminal_twoFold`]
For a partial refinement pr *of length L with σ injective, every $t \in \mathbf{I}$ lies in at most two of $\overline{J_1}, \dots, \overline{J_L}$.*

[2] A variant `step_succ` chooses i^* internally via Finset.min′; `step_succ_at` takes i^* as an explicit argument and returns the extension bundled with the "i^* is covered" property, for termination.

Argument. From spacing condition (a), iterated m times, one obtains that indices i, j with $|i - j| = 2(m+1)$ (an even gap ≥ 2) have strict separation between $\overline{I_i}$ and $\overline{I_j}$, hence $\overline{I_i} \cap \overline{I_j} = \emptyset$ (✓ `chain_spacing`, ✓ `closure_disjoint_of_even_gap`).

Suppose three distinct indices $k_1, k_2, k_3 \in \{1, \ldots, L\}$ admit $t \in \overline{J_{k_1}} \cap \overline{J_{k_2}} \cap \overline{J_{k_3}}$. Since $\overline{J_k} \subseteq \overline{I_{\sigma(k)}}$ and σ is injective, $\sigma(k_1), \sigma(k_2), \sigma(k_3)$ are three distinct indices in $\{1, \ldots, n\}$ with a common point in the closures of $I_{\sigma(k_i)}$. Among any three distinct integers $a < b < c$ some pair has even gap ≥ 2: either $c - a$ is even (and necessarily ≥ 2), or $c - a$ is odd and ≥ 3, in which case $b - a$ and $c - b$ have opposite parities and one of them is the even gap. The pair with even gap has disjoint closures, contradicting the common point (✓ `not_three_overlap`). □

Proposition 3.6 (Closure extension of saving bound)**.** [✓ `saving_bound_closure`]

Let pr *be a partial refinement. For every k and every $t \in \overline{J_k}$, $f(t) - E_{\sigma(k)}(t) \geq 1/(4N)$, where $E_{\sigma(k)}$ is the replacement energy of the $\sigma(k)$-th local witness.*

Argument. On $J_k \subseteq U_{w_{\sigma(k)}}$ the witness gives $f - E_{\sigma(k)} \geq 1/(4N)$. The function $g := f - E_{\sigma(k)}$ is continuous (Definitions 2.2 requires E_t continuous; f is continuous by hypothesis). A closure point is a limit of a sequence in J_k, and g preserves $\geq$-inequalities under limits. □

Lemma 3.7 (Assembly)**.** [✓ `exists_refinement`]

A partial refinement pr *of length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i yields a refinement of $(\mathbf{I}, X, f, m_0, N)$ in the sense of Definition 2.4, with $p_k := \overline{J_k}$, uniform saving $s_k := 1/(4N)$, and $E_k :=$ the replacement energy of the $\sigma(k)$-th witness.*

Argument. Invariant (I) is Proposition 3.6. Invariant (II) is Proposition 3.5. Invariant (III) holds by construction ($s_k = 1/(4N)$). Coverage of NC chains $\mathrm{NC} \subseteq \bigcup_i I_i \subseteq \bigcup_k J_k \subseteq \bigcup_k \overline{J_k}$. □

Theorem 2.8 now follows: chain Propositions 3.1 and 3.4, invoke Lemma 3.7, and apply Theorem 2.7.

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